



Welcome to the first edition of the AI4Agri Task!

Motivation

The AI4Agri task is motivated by the pressing need to enhance sustainable and efficient agricultural practices through advanced artificial intelligence. This initiative is structured around two critical subtasks, each with distinct geographical data sources: **AgriPotential**, which focuses on predicting land suitability for various crop types using multi-spectral Sentinel-2 imagery from Southern France for sustainable planning; and the **DACIA5**, which specifically tackles Crop identification/Early crop identification by using its multi-source dataset (Sentinel-1 SAR and Sentinel-2 MSI) from a specific area in Braşov, Romania to accurately identify diverse crop types early in their growth cycle. Together, these efforts leverage AI to optimize agricultural potential and enable more informed decision-making.

Task Description

The AI4Agri Task is structured into two distinct yet complementary sub-tasks designed to advance agricultural intelligence. Participants are free to choose any (or both) of the subtasks.

Quick navigation:

- [Subtask 1](#)
- [Subtask 2](#)

Subtask 1: AgriPotential: Viticulture Edition

Motivation

Remote sensing has emerged as a critical tool for large-scale Earth monitoring and land management. A novel task within this domain is the **estimation of agricultural potentials** - often referred to as agricultural productivity, soil capability, capacity, or crop suitability - which describe the ability of a specific area to support agricultural production. Unlike classical remote sensing problems in smart agriculture, which focus on estimating yields and mapping crops during or after the growing season, predicting agricultural potentials is particularly valuable before planting. It allows the assessment of land suitability regardless of the current land use, making it important for early decision-making. Traditionally, agricultural potentials are assessed using manual field experimentation and in-situ measurement performed by domain experts. These approaches are time-consuming and labor-intensive, and require frequent repetition to remain up-to-date. Exploiting remote sensing data in order to provide a scalable, timely, alternative for assessing agricultural potentials could be an alternative worth examining. From a technical point of view, this challenge motivates the use of multi-spectral time series modeling and ordinal modeling.

Task description

The participants will be challenged to build a model or a pipeline capable of estimating agricultural potentials specifically for viticulture (grapevine cultivation) from multi-temporal multi-spectral satellite imagery. The goal is to predict the potential classes that range from very low to very high on an ordinal scale.

We encourage participants to refer to

<https://www.codabench.org/competitions/12055/> for more detailed information.

Important dates

- **Submissions Start: December 15, 2025**
- **Participant Registration Closes - April 23, 2026**
- **Submissions Close: May 7, 2026**
- **[CEUR WS] Notebook paper submission (report): May 28, 2026**
- **[CEUR WS] Acceptance notification: June 30, 2026**
- **[CEUR WS] Notebook camera-ready: July 6, 2026**
- **CLEF 2026 Conference in Jena, Germany : 21-24 September 2026**

Satellite Image Time Series

Satellite Image Time Series (SITS) refers to a sequence of satellite images of the same geographic area captured over time. In this challenge, multi-spectral SITS are used in agricultural monitoring.

A **multi-spectral image** captures light reflected from the Earth's surface in multiple spectral bands, which are specific ranges of wavelengths of light. Unlike regular images that only show visible light (i.e red, green, blue (RGB)) multi-spectral images can also include light that humans cannot see, such as infrared.

The images used in this challenge come from **Sentinel-2**, a satellite operated by the European Space Agency (ESA). Sentinel-2 provides high-resolution imagery in 13 spectral bands and revisits the same location frequently (every 5 days). This makes it ideal for creating time series that capture temporal variations of the Earth.

In general, one major issue in optical STIS is cloud cover. Clouds block that satellite's view of the land and some images may be obscured and unusable. Because of clouds, the time series can be irregular, meaning that the images are not acquired at evenly spaced intervals. However, in Agripotential, the global cloud cover is <2%, so it has a minimal impact and does not significantly affect analysis.

Data

The dataset that is used in this challenge consists of:

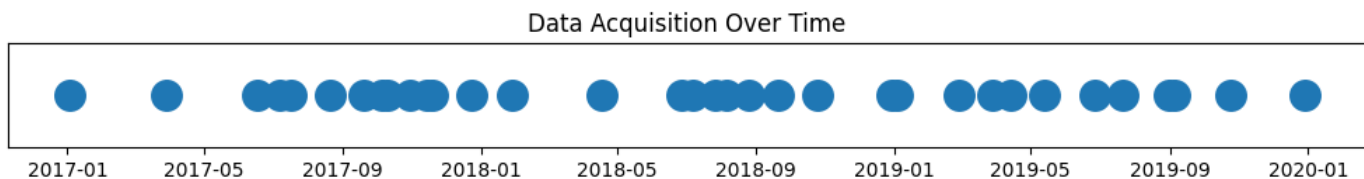
- 34 Sentinel-2 timeframes collected between 2017 and 2019. Each image has 10 spectral bands
- Ground truth labels representing the 5 possible classes of viticulture potential. The labels are annotated on the pixel-level

The dataset is based on the AgriPotential benchmark dataset that is presented in

<https://arxiv.org/pdf/2506.11740>.

Unlike the original version of **AgriPotential**, the images of the challenge are not normalized, leaving the choice of normalization techniques to the users.

The data is acquired at irregular intervals in time. The figure below shows the acquisition dates for the 34 Sentinel-2 timeframes from 2017 to 2019.



The data is accessible in three ways:

- Offline, by downloading the dataset (~200GB) from huggingface:
<https://huggingface.co/datasets/m-sakka/agripotential/tree/main>
- Offline, by downloading the dataset using the Agripotential python package:
<https://github.com/MohammadElSakka/agripotential>
- Online, by using the same Agripotential python package

A tutorial on GitHub is provided on how to load the data, visualize it, and make it usable for practicing: <https://github.com/MohammadElSakka/agripotential>

The provided data contains two subsets:

- A training subset of 6329 patches*
- A validation subset of 781 patches
- A test subset** of 800 patches

* Each patch is a satellite time series of 34 timeframes and 10 spectral bands.

** Ground truth labels are not provided for the test set.

The table below summarizes the dataset main properties:

Property	Value
Dataset name	AgriPotential
Data volume	200 GB
Timeframes	34
Spectral channels	10
Spatial resolution	5 m/px
Annotation level	Pixel level
Potential classes	Very low, low, average, high, very high
License	CC BY 4.0
Data link	https://huggingface.co/datasets/m-sakka/agripotential/tree/main

Evaluation

Agricultural potential are represented by ordered classes from “very low” to “very high”. The evaluation will be conducted using the following metric:

Accuracy±1: known as “Accuracy with ±1 tolerance”, which measures the proportion of predictions that are within one class of the true label. The ordinal nature of the classes allows predictions that are close to the true class to be considered partially correct.

Participant guidelines

Please refer to the general [ImageCLEF registration instructions](#)

Note: Thie task does not use the Ai4Media benchmarking platform for evaluation. Creating an account on the platform for this task is not required. Completing the registration form is still required, similar to the other tasks. You can also find the form [here](#).

All entries to this competition **must** be submitted directly through **CodaBench** on the following link : <https://www.codabench.org/competitions/12055/>. Participants are required to submit their model **outputs** within a **ZIP file**. All output files must be placed directly in the root of the ZIP archive. Further instructions and a complete walkthrough of the submission processes are provided on Codabench. In case you encounter any problems with registering please contact us *immediately* to find a solution.

The process of evaluating submissions and ranking is handled by the Codabench platform.

Prizes

The best submission will be granted a registration at the CBMI 2026 conference that will be held in Toulouse, France in October 2026. More information will be communicated later.

Reference (accepted at CBMI 2025)

@article{sakka2025agripotential,
title={AgriPotential: A Novel Multi-Spectral and Multi-Temporal Remote Sensing Dataset for

```
Agricultural Potentials}},
author={Sakka, Mohammad El and De Pourtales, Caroline and Chaari, Lotfi and Mothe, Josiane},
journal={arXiv preprint arXiv:2506.11740},
year={2025}
}
```

Contact

- o Mohammad El Sakka: mohammad.el-sakka@outlook.com

Subtask 2: DACIA5

Focusing on remote sensing for agricultural progress, this sub-task leverages advanced satellite data to drive innovation in farming practices. This subtask encourages innovative uses of the DACIA5 dataset, including the development of new models, integration through data aggregation or fusion with complementary sources, and the exploration of novel research directions beyond the scope of the original publication.

DACIA5 sub-task is divided into two main challenges:

1. Crop identification: Past_vs_present

For the first challenge, we have defined the "present" moment as the beginning of 2024. Consequently, all prior data (from 2020, 2021, 2022, and 2023) is considered available for training, representing the "past". In contrast, all data from 2024 is allocated to the testing set, simulating "present" and unseen data.

2. Early crop identification

This challenge focuses on the timely identification of specific crops, namely "winter wheat" and "alfalfa", during the early spring season (i.e., March). Participants will train their models on a comprehensive dataset, carefully excluding the March patches of these two specific crops from 2022, 2023, and 2024. These excluded March patches then form the testing set, allowing for evaluation of the models' ability to accurately identify "winter wheat" and "alfalfa" at a crucial early stage of their growth cycle.

Data

Subtask 2: DACIA5

The dataset provided for this subtask is the DACIA5. The dataset is publicly available at [Zenodo](#) and was originally introduced in the publication: "[DACIA5: a Sentinel-1 and Sentinel-2 dataset for agricultural crop identifications](#)".

The dataset contains all the available images from Sentinel-1 SAR and Sentinel-2 MSI from 2020-2024 over a specific area to the north of Braşov city, Romania, together with the 32 x 32 pixel radar and multi-spectral patches for a crop identification task using a learning model.

- o **172 Sentinel-2 images:** Each image has a dimension of 800 x 450 x 12 (height x width x spectral bands) and is saved in GeoTIFF format;
- o **159 Sentinel-1 images:** Each image has a dimension of 800 x 450 x 2 (height x width x radar channels) and is saved in GeoTIFF format. For each Sentinel-2 image we have a corresponding Sentinel-1 image; this correspondence was made based on the acquisition data (a Sentinel-1 image may correspond to multiple Sentinel-2 images);
- o **17 crop types;**
- o **6454 optical patches:** 32 x 32 x 12 (Sentinel-2 patches) saved in GeoTIFF and MAT format;
- o **5995 radar patches:** 32 x 32 x 2 (Sentinel-1 patches) saved in GeoTIFF and MAT format;
- o For each year, there is a mask showing all the parcels and the crops from the [National Institute of Research and Development for Potato and Sugar Beet \(NIRDPSB\)](#).

The table below summarizes the dataset main properties:

Property	Value
Dataset name	DACIA5
File format	GeoTIFF (.tif) and .mat
File size	3.4 GB
Number of optical images	172 (Sentinel-2)
Number of radar images	159 (Sentinel-1)
Number of optical patches (32 x 32)	6454 (Sentinel-2)
Number of radar patches (32 x 32)	5995 (Sentinel-1)
Timestamps	331
Spectral bands (Sentinel-2)	12
Radar channels (Sentinel-1)	2
Spatial resolution	10 meters
Dimensions of images (height x width)	800 x 450
Dimensions of patches (height x width)	32 x 32
Annotation level	Pixel level
Crop types	agriculture crops
Potential classes	winter wheat, corn, corn silage, peas, winter rapeseed, late potato, other potato, spring wheat, soybean, sugar beet, alfalfa
License	CC BY 4.0
Data link	https://zenodo.org/records/14283243

Evaluation Methodology

Subtask 2: DACIA5

Crop identification: Past_vs_present

Evaluation details:

- Input data is given as (32 x 32) pixels x 12 bands patches. For each patch, the model must predict one of the classes: "Wheat":0, "Corn":1, "Peas":2, "Rapeseed":3, "Potato":4, "Sugarbeet":5, "Alfalfa":6
- Note that some of the sought classes are made from merging original DACIA5 ones:
 - "Wheat" = "winter wheat" + "spring wheat"
 - "Corn" = "Corn" + "Corn silage"
 - "Potato" = "Late potato" + "other potato"
- The "Soybean" class existing in the DACIA5 dataset is not included in the testing set.

Quality metric: $Q_1 = 0.5 \cdot AA + 0.5 \cdot OA$

Where:

- AA** is the **Average Accuracy**: The accuracy to recognize each class, that is further averaged

between all 7 classes. The value is given as a percentage, thus between 0 and 100.

- **OA is Overall Accuracy:** is the accuracy counted over the entire testing set, without taking into account the crop type. The value is given as a percentage, thus between 0 and 100.
- The resulting quality metric, Q_1 is, thus, up to 100.

Note: Participants are encouraged to divide the training set into "Training" and "Validation".

Early crop identification

Evaluation details:

- Input data is given as (32 x 32) pixels x 12 bands patches. For each patch the model must predict one of two classes: "Winter Wheat":0, "Alfalfa":1

Quality metric: $Q_2 = 0.5 \cdot \text{Acc}_{\text{spring_wheat}} + 0.5 \cdot \text{Acc}_{\text{alfalfa}}$

Note: Taken into account that the amount of data for these specific two classes is limited, we suggest to build a deep model for embedding separation (clustering), followed by nearest neighbor classifier.

Submission Guidelines and Instructions

Subtask 2: DACIA5

To foster innovation and collaboration, the DACIA5 subtask is open to a wide range of early-career researchers, including undergraduate, Master's, and PhD students, as well as postdoctoral researchers and junior research assistants/fellows. While encouraging young talent, each team also has the option to be supervised by a senior researcher. Participants from academia, research institutes, and independent backgrounds are all warmly welcomed to contribute their expertise.

1. Code

- The implementation should be shared either as a Google Colab notebook, or as a zipped folder containing the source code and clear instructions for execution (README file).
- All dependencies must be clearly listed. Use of pre-trained models or open-source libraries is allowed and encouraged, as long as it is clearly documented.
- The first three teams for each track must share, also, the training code, which needs to be replicable, under reasonable assumptions.

2. Technical report (max. 3 pages)

A short technical report or extended abstract summarizing:

- The approach taken (modeling strategy, data preprocessing, etc.)
- The track or direction explored (e.g., performance improvement, dataset fusion, transfer learning, etc.)
- The main results
- [Optional] observations, limitations, or suggestions for future work

3. [Optional] Supplementary materials

- Visualizations, tables, or diagrams that help explain your solution are welcome.
- A brief video pitch (max. 2 min) is optional but can strengthen the submission.

All materials should be submitted via ai4agri.imageclef2026@gmail.com

Please ensure that:

- The email subject line or file names clearly include:
 - The participant's name(s)
 - The challenge: This can be either "Crop identification: Past_vs_present" or "Early crop identification"

- Teams that want to submit to both tracks are kindly asked to make separate submissions.

Participant Registration

Please refer to the general [ImageCLEF registration instructions](#)

Note: This task does not use the Ai4Media benchmarking platform for evaluation. Creating an account on the platform for this task is not required. Completing the registration form is still required, similar to the other tasks. You can also find the form [here](#).

CEUR Working Notes

Citations

Subtask 2: DACIA5

When referring to the training data please use the following citation:

Ivanovici, M., Baicoianu, A., Plajer, I. C., Debu, M., Ștefan, F.-M., Florea, C., Cațaron, A., Coliban, R.-M., Popa, S., Oprisescu, S., Racoviteanu, A., Olteanu, G., Marandskiy, K., Ghinea, A., Kazak, A., Majercsik, L., Manea, A., & Dogaru, L. (2024). *AI4AGRI Sentinel-2 Brasov area 2020-2024 multi-spectral dataset for crop monitoring and identification* [Data set]. Zenodo.

<https://doi.org/10.5281/zenodo.14283243>

```
@dataset{ivanovici_2024_14283243,
  author = {
    Ivanovici, Mihai and
    Baicoianu, Alexandra and
    Plajer, Ioana Cristina and
    Debu, Matei and
    Ștefan, Floriana-Maria and
    Florea, Corneliu and
    Cațaron, Angel and
    Coliban, Radu-Mihai and
    Popa, Stefan and
    Oprisescu, Serban and
    Racoviteanu, Andrei and
    Olteanu, Gheorghe and
    Marandskiy, Kamal and
    Ghinea, Adrian and
    Kazak, Artur and
    Majercsik, Luciana and
    Manea, Adrian and
    Dogaru, Liviu
  },
  title = {
    AI4AGRI Sentinel-2 Brasov area 2020-2024 multi-
    spectral dataset for crop monitoring and
    identification
  },
  month = dec,
  year = 2024,
  publisher = {Zenodo},
  doi = {10.5281/zenodo.14283243},
  url = {https://doi.org/10.5281/zenodo.14283243},
}
```

Contact

Subtask 2: DACIA5

- Alexandra Băicoianu: a.baicoianu@unitbv.ro, Braşov, Romania
- Corneliu-Nicolae Florea: corneliu.florea@upb.ro, Braşov, Romania
- Mihai Ivanovici: mihai.ivanovici@unitbv.ro, Braşov, Romania

Acknowledgments

For the AI4Agriculture task, the creation of the DACIAS dataset was funded by the European Union through the AI4AGRI project (Romanian Excellence Center on Artificial Intelligence on Earth Observation Data for Agriculture) which received funding from the EU's Horizon Europe research and innovation programme under the grant agreement no. 101079136. The Agripotential dataset was also partially funded by that project.

Attachment	Size
 S2 dates	11.23 KB